

Module 23.10: Understanding the Cost Function Curve (Convexity & Contour Plots)

1. Feature Space vs. Parameter Space

Understanding machine learning requires switching between two complementary geometric representations:

Dimension	Feature Space (Data Space)	Parameter Space (Loss Surface)
Axes	Horizontal: x , Vertical: y	Horizontal: w , Vertical: b (and Height: J)
Data Points	Individual scatter points $(x^{(i)}, y^{(i)})$	Transformed into constants within the cost equation
Model	A candidate straight line $\hat{y} = wx + b$	A single point (w, b) on the landscape
Goal	Find line that fits data scatter	Find the lowest coordinate (valley bottom) in the landscape

2. The 1D Cost Curve: A Quadratic Parabola

To understand the geometry, consider the simplified univariate case where intercept $b = 0$. The hypothesis is $\hat{y} = wx$.

$$J(w) = \frac{1}{2m} \sum_{i=1}^m (wx^{(i)} - y^{(i)})^2 = \frac{1}{2m} \sum_{i=1}^m \left[(x^{(i)})^2 w^2 - 2x^{(i)}y^{(i)}w + (y^{(i)})^2 \right]$$

Rearranging terms in powers of w :

$$J(w) = Aw^2 - Bw + C$$

Where:

$$A = \frac{1}{2m} \sum_{i=1}^m (x^{(i)})^2, \quad B = \frac{1}{m} \sum_{i=1}^m x^{(i)} y^{(i)}, \quad C = \frac{1}{2m} \sum_{i=1}^m (y^{(i)})^2$$

Since $(x^{(i)})^2 \geq 0$, coefficient $A > 0$ for any non-trivial dataset. Therefore, $J(w)$ is a **parabola opening upwards** (convex quadratic curve). It has a single unique global minimum at the vertex:

$$w^* = \frac{B}{2A} = \frac{\sum x^{(i)} y^{(i)}}{\sum (x^{(i)})^2}$$

3. The 2D Cost Landscape: 3D Paraboloid Bowl

When both parameters w and b are variable, the cost function $J(w, b)$ forms a 3-dimensional surface shaped like a **bowl** (an elliptic paraboloid).



Strict Convexity and the Hessian Matrix

A function is strictly convex if its Hessian matrix $\mathbf{H}$ (matrix of second-order partial derivatives) is positive definite everywhere:

$$\mathbf{H} = \begin{bmatrix} \frac{\partial^2 J}{\partial w^2} & \frac{\partial^2 J}{\partial w \partial b} & \frac{\partial^2 J}{\partial b \partial w} & \frac{\partial^2 J}{\partial b^2} \end{bmatrix} = \begin{bmatrix} \frac{1}{m} \sum (x^{(i)})^2 & \frac{1}{m} \sum x^{(i)} & \frac{1}{m} \sum x^{(i)} & 1 \end{bmatrix}$$

Evaluating the determinant of $\mathbf{H}$:

$$\det(\mathbf{H}) = \frac{1}{m} \sum (x^{(i)})^2 - (\bar{x})^2 = \text{Var}(X)$$

Because sample variance $\text{Var}(X) > 0$ whenever x contains varying values, $\mathbf{H}$ is strictly positive definite.

Key Consequence: Linear regression with MSE has **no local minima**! Any local minimum is guaranteed to be the unique **global minimum**.

4. Contour Plots (Level Curves)

A **contour plot** projects 3D surface slices of constant cost $J(w, b) = c$ onto a 2D plane:

- Each closed elliptical ring represents a set of parameters (w, b) yielding identical cost.
- The concentric rings contract toward a single central point: the global minimum (w^*, b^*) .
- The gradient vector $\nabla J(w, b)$ is always **strictly perpendicular** to the tangent of the contour curve at that point.